Information Retrieval

Introduction

IR models and methods

Search and Information Retrieval

- Search on the Web is a daily activity for many people throughout the world
- Search and communication are the most popular uses of the computer
- Applications involving **search** are **everywhere**

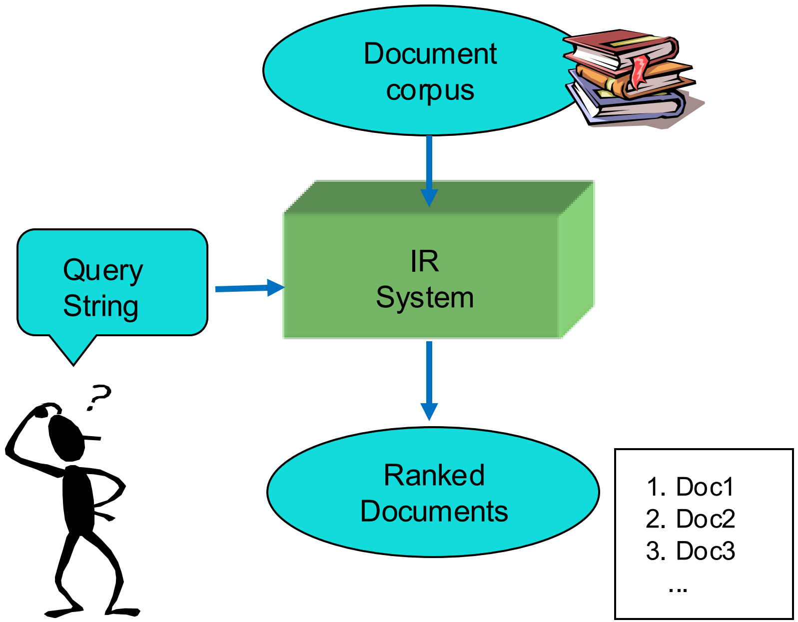
- The field of computer science that is most involved with R&D for search is **information retrieval (IR)**

Information Retrieval

- *"Information Retrieval (IR)
  is* ***finding material*** *(usually documents)
  of an* ***unstructured*** *nature (usually text)
  that satisfies an* ***information need***
  from within ***large collections*** *(usually stored on computers)."*
  (Manning, et al, 2008)

Typical IR Task

- Given:
  - A set of documents (corpus)
  - A user query in the form of a textual string
- Find:
  - A ranked set of documents with information that is relevant to the user's information need and helps the user complete a task



What is a Document?

- Examples:
  - web pages, emails, books, news stories, scholarly papers, text messages, forum postings, patents, social media, etc.
  - TXT, PDF, HTML, JSON, XML, ..etc.

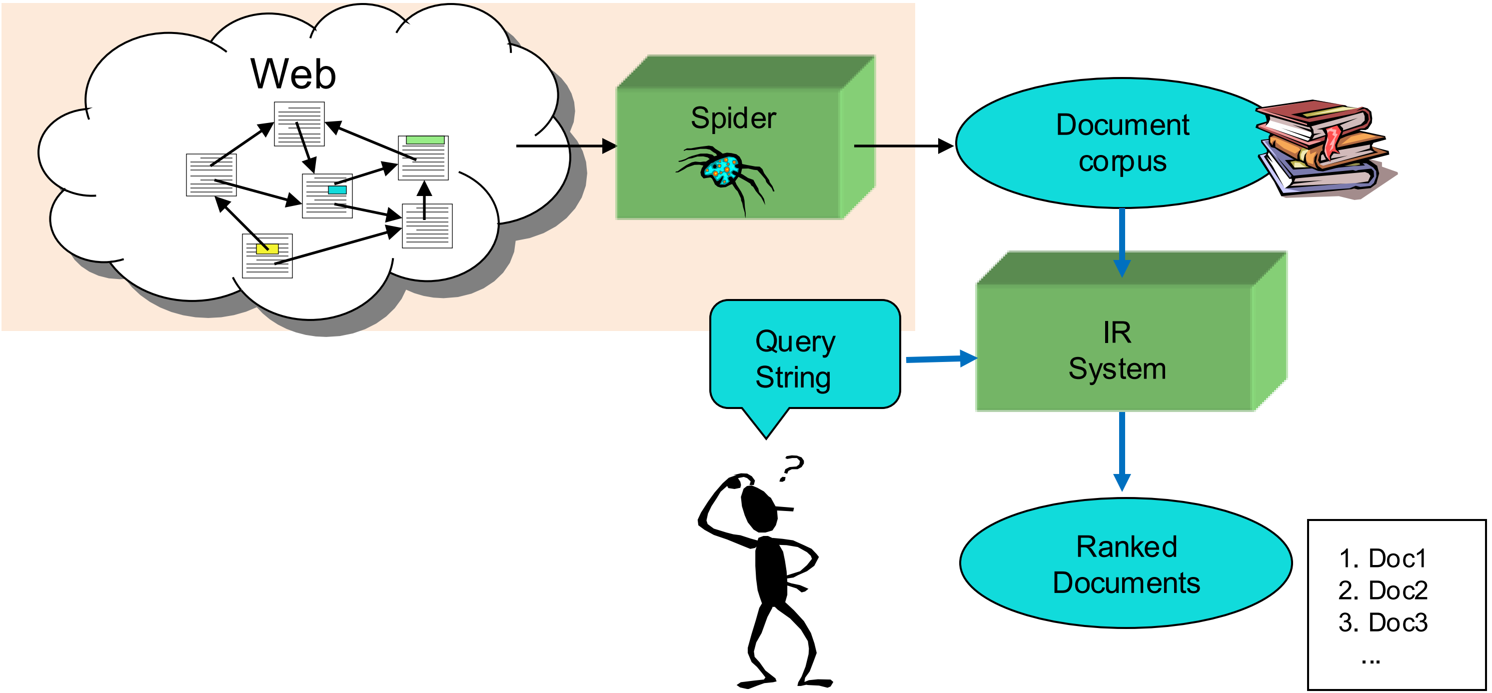
- Common properties
  - Significant text content
  - Some structure
    - e.g.,
    - title, author, and date, for papers
    - subject, sender, and destination, for emails

Web Search

- Application of IR to HTML documents

- Differences:
  - Must assemble document corpus by spidering the web
  - Can exploit the structural layout information in HTML
  - Can exploit the link structure of the web
  - Documents change uncontrollably

Web Search


Web
Spider
Document
corpus
Query
String
IR
System
Ranked
Documents
1. Doc1
2. Doc2
3. Doc3
...

IR – Related Areas

- Database Management
  - Easy to compare fields with well-defined semantics to queries in order to find matches
- Library and Information Science
  - Digital libraries
- Artificial Intelligence
  - Web ontologies and intelligent information agents
- Natural Language Processing
  - Analyzing syntax (phrase structure) and semantics
  - Disambiguation
- Machine Learning
  - Classification / Clustering
  - Learning to rank
- Representation Learning and Deep Learning
  - Neural language models
  - Can combine Machine Learning with Semantics (and syntax)

Big Issues in IR

- Relevance
- Evaluation
- Information needs

Big Issues in IR - Relevance

- What is it?
  - Simple (and simplistic) definition: A relevant document contains the information that a person was looking for when they submitted a query to the search engine
- Many factors influence a person's decision about what is relevant: e.g., task, context, novelty, style
- Topical relevance (same topic) vs. user relevance

Big Issues in IR - Relevance

- Ranking algorithms used in search engines are based on retrieval models
- Each retrieval model defines a view of relevance
- Most models describe statistical properties of text rather than linguistic
  - i.e. counting simple text features such as words instead of parsing and analyzing the sentences
  - Statistical approach to text processing started with Luhn in the 50s
  - Linguistic features can be part of a statistical model

Notion of relevance –> IR Models

- Boolean model
- Vector-space model
- Probabilistic model
- Language-based model
- Neural model

Big Issues in IR - Evaluation

- Experimental procedures and measures for comparing system output with user expectations
  - Originated in Cranfield experiments in the 60s
- IR evaluation methods now used in many fields
- Typically use test collection of documents, queries, and relevance judgments
  - Most commonly used are TREC collections
- Recall and precision are two examples of effectiveness measures

Big Issues in IR - Evaluation

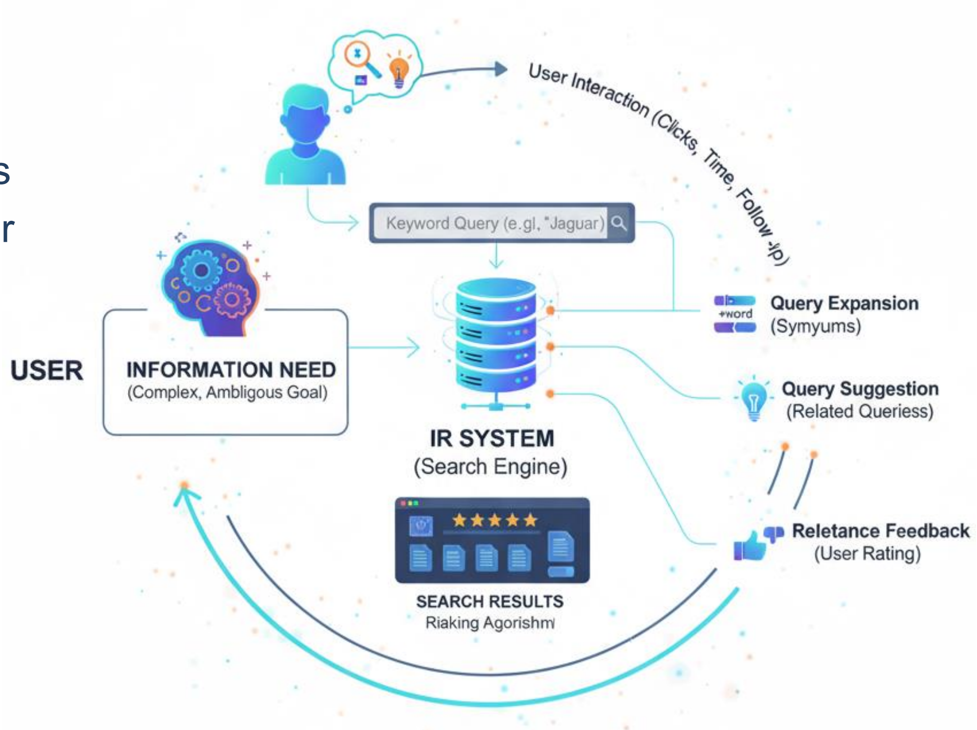
- How good are the retrieved docs?
  - *Precision* : Fraction of retrieved docs that are relevant to the user's information need
  - *Recall* : Fraction of relevant docs in collection that are retrieved

  - We will look at more precise definitions and measurements later

Big Issues in IR - Information Needs

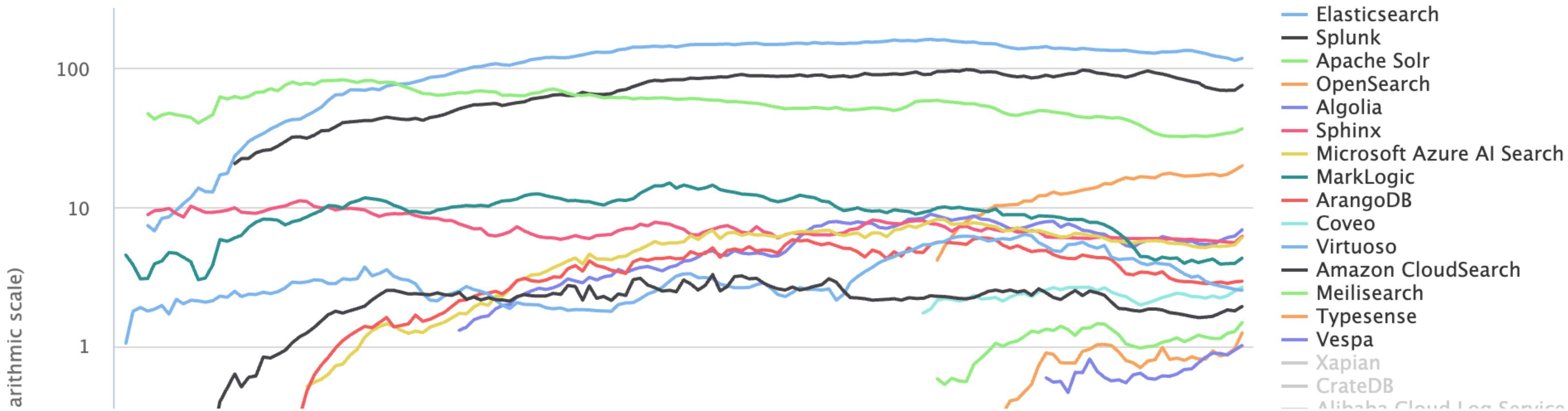
- Search evaluation should be user-centered
- Keyword queries are often poor descriptions of actual information needs
- Interaction and context are important for understanding user intent

- Query refinement techniques improve ranking
  – *query expansion, query suggestion, relevance feedback*



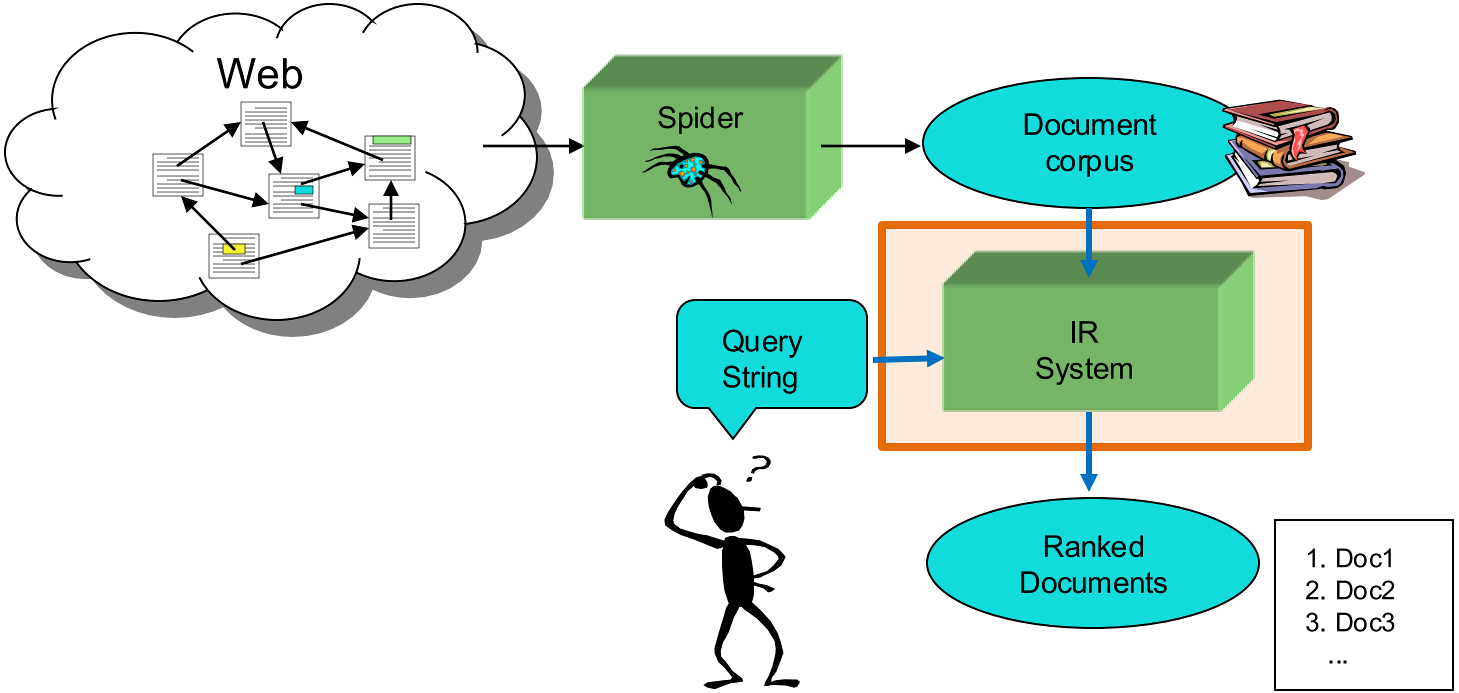
IR and Search Engines

- Web search engines are the best-known examples, but many others (e.g. enterprise search)
  - But open-source search engines are important for research and development
    - e.g., Elasticsearch, Apache Solr, Apache Lucene, Sphinx, …

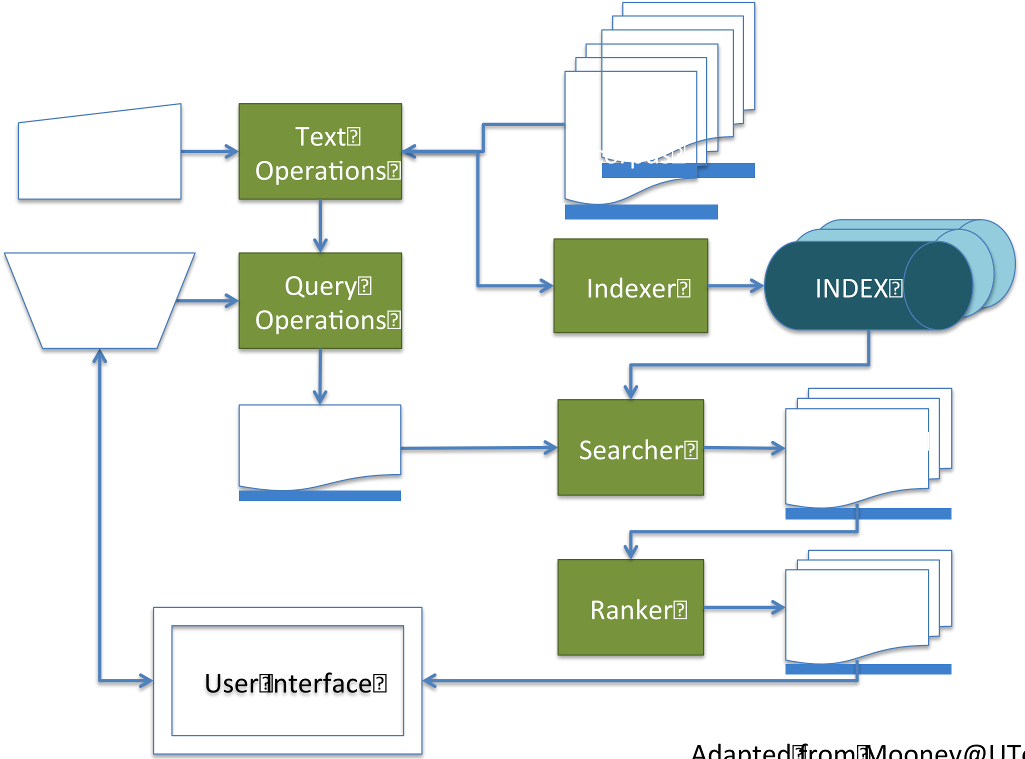


https://db-engines.com/en/ranking_trend/search+engine

IR System Architecture

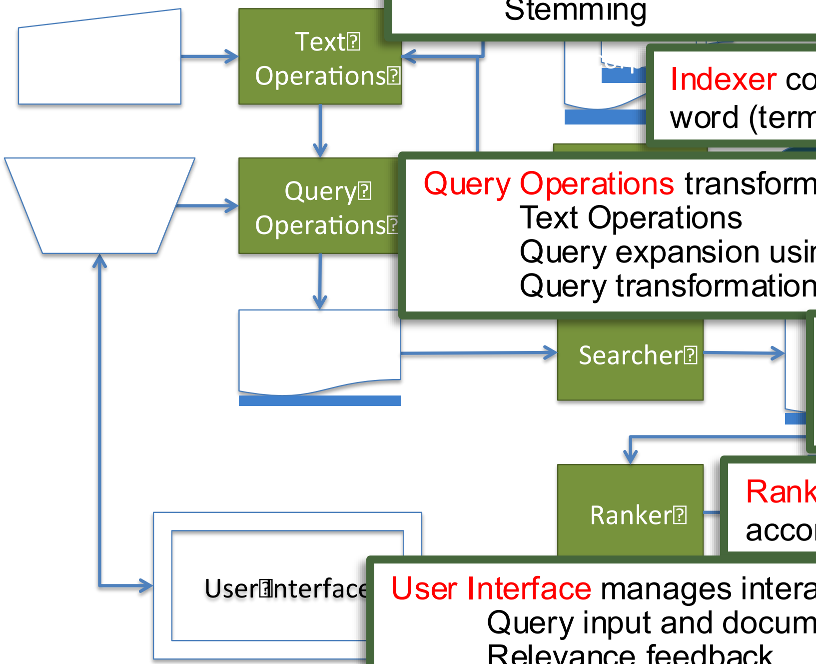


IR System Architecture



Adapted from Mooney@UTexas

IR System Architecture



Text Operations forms index words (terms)
Stopword removal
Stemming

Indexer constructs an inverted index of word (term) to document pointers.

Query Operations transform the query to improve retrieval:
Text Operations
Query expansion using a thesaurus.
Query transformation using relevance feedback.

Searcher retrieves documents that contain a given query term from the inverted index.

Ranker scores all retrieved documents according to a relevance metric.

User Interface manages interaction with the user:
Query input and document output.
Relevance feedback.
Visualization of results.

Key Terms Used in IR

- **QUERY**
  - a representation of what the user is looking for - can be a list of words or a phrase.
- **DOCUMENT**
  - an information entity that the user wants to retrieve
- **COLLECTION**
  - a set of documents
- **INDEX**
  - a representation of information that makes querying easier
- **TERM**
  - word or concept that appears in a document or a query

Other Important Terms

- Classification
- Cluster
- Similarity
- Information Extraction
- Term Frequency
- Inverse Document Frequency
- Precision
- Recall
- Inverted File
- Query Expansion
- Relevance
- Relevance Feedback
- Stemming
- Stopword
- Vector Space Model
- Weighting
- TREC/TIPSTER/MUC